

Code-Layout, Readability and Reusability

Date	1 July 2026
Team ID	XXXXXX
Project Name	HouseRent – House Rental Management System
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing and indentation are maintained throughout the codebase.	Yes	Improves readability and consistency.
2	Proper File Structure	Client and Server are organized into separate folders with logical project structure.	Yes	Follows MERN project architecture.
3	Meaningful Variable Names	Variables and constants use descriptive names that clearly indicate their purpose.	Yes	Easy to understand and maintain.
4	Function / Method Names	Functions and methods are named based on their functionality.	Yes	Improves code readability.
5	Code Comments	Important logic and complex sections are documented using comments.	Partial	Comments are added where necessary.

6	Modular Design	Code is divided into reusable components, routes, models, and middleware.	Yes	Promotes reusability and maintainability.
7	No Redundant Code	Duplicate and unnecessary code has been minimized.	Yes	Reusable functions and components are used.
8	Error Handling	API errors and exceptions are handled appropriately using Express middleware and validations.	Yes	Provides better user experience and debugging.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
1	Navbar Component	React.js	All Client Pages	High
2	Authentication Middleware	Node.js / Express.js	Protected API Routes	High
3	User Model	MongoDB / Mongoose	Authentication & User Management	High
4	Property Model	MongoDB / Mongoose	Property Management & Booking	High
5	Booking Routes	Express.js	Tenant & Owner Modules	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized MERN project structure with separate Client and Server modules.
Readability	5	Descriptive naming conventions and consistent formatting improve readability.
Reusability	5	Modular React components, Express routes, middleware, and database models are reusable.
Documentation / Comments	4	Important logic is documented; additional comments can be added for complex sections.
Overall Score	4.8 / 5	The project follows good coding standards, modular architecture, and maintainable design practices.

